

EXPERIENCE

- AI Research Engineer

January 2026 - Present

Metafore

Bengaluru

– Built an end-to-end fine-tuning platform spanning dataset preparation, model training, deployment, and downstream delivery.

– Designed LLM-agent workflows to generate, validate, and reconcile ontology mappings across enterprise schemas.

– Developed a UI-driven agent generator that scaffolds production agents from Business Requirements Documents.

– Architected MetaFabric as an AI-assisted ontology platform, mapping disparate ITSM data models via TTL files to a unified canonical model exposed through developer-friendly APIs.
- Software Engineering Consultant

August 2025 - Jan 2026

Assured Health Technologies

Bengaluru

– Executed 17 Python data-migration scripts across the production database to update multiple application fields.

– Automated AWS deployments and operations with CloudFormation, decreasing overall end-to-end deployment time by 67%.

– Cut Django API latency from 35.7s to 3.7s using database indexes and eliminating redundant mutational prefetch queries.

– Adopted the Bun JavaScript runtime across application workflows, reducing combined compilation and runtime by 53%.
- Software Engineering Intern

July 2024 - September 2024

CNCF / Litmus Chaos

Remote

– Created a Litmus upgrade agent that streamlined version transitions, cutting manual intervention by 30% during upgrades.

– Orchestrated sequential upgrade pipelines that reduced version-migration time for control-plane and Litmus database updates.

– Packaged MongoDB schema migrations for every Litmus version through release-specific deployment manifests.

– Optimized the Auth server login API by 98.84%, reducing measured login latency from 1.73 seconds to 15.87 milliseconds.
- Software Engineering Intern

April 2023 - July 2023

Organic Maps

Remote

– Deployed a TypeScript/NodeJS proxy serving 50K+ daily requests at 99.9% uptime on Cloudflare Workers.

– Converted Google Maps URIs into mobile-ready custom formats, reducing application load time by 60% for mobile clients.

– Connected the proxy across iOS and Android clients via streamlined API calls, improving mobile data-fetch speed by 30%.

PERSONAL PROJECTS

- LLM Inference Engine

A from-scratch LLM inference engine in PyTorch (KV cache, batching, paged memory)

Github

– Implemented a PyTorch GPT-2/Mistral engine with KV caching, continuous batching, paged memory, and async APIs.

– Increased decode throughput 2.22x and cut CUDA execution time 46.6% using a preallocated KV cache on one A100 GPU.

– Scaled GPT-2 124M throughput from 155 to 18,346 tokens/second, a 118.3x gain at batch size 512 on one A100 GPU.

– Reduced KV-cache memory 35.9%, reached 3.2x throughput at 64 concurrent requests, and validated behavior with 122 tests.
- Kimi-K3-Impl

Reproduction of a 49M-parameter Kimi K3-inspired language model, trained on a single 8GB GPU

Github

– Reproduced a 49M-parameter Kimi K3 model with KDA, Gated MLA, Attention Residuals, and Stable LatentMoE layers.

– Trained on TinyStories in BF16 on one RTX 4060 Laptop GPU; selected final checkpoint 9,750 at 1.6277 validation loss.

– Integrated Triton KDA kernels, cached KDA/MLA autoregressive decoding, Muon optimizers, and an FP32 CPU fallback.

– Added atomic checkpoints, live training metrics, and 6 tests covering architecture, causality, decoding, and artifact integrity.

TECHNICAL SKILLS AND INTERESTS

AI/ML Systems: PyTorch, Triton, CUDA, LLM inference, fine-tuning, KV caching, continuous batching

Languages: Python, Go, TypeScript, JavaScript, Rust, Java, C++, SQL

Infrastructure/Data: AWS, Docker, Kubernetes, CloudFormation, Prometheus, OpenTelemetry, Kafka, MongoDB, MySQL

EDUCATION

- Indian Institute of Technology Patna

2025

Bachelor of Technology